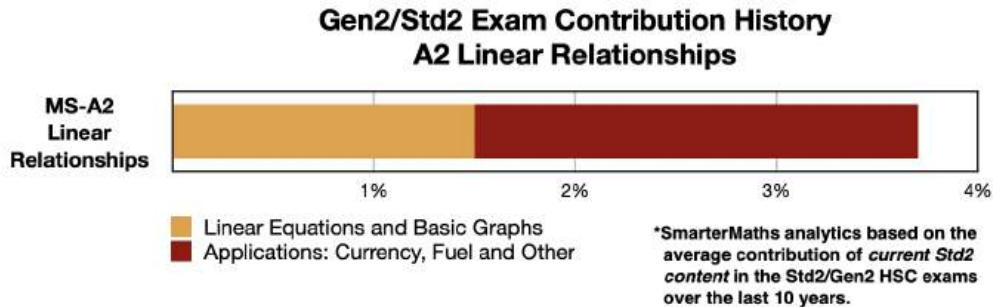


STD 1: Algebra (Std 1), A2 Linear Relationships (Y11)
Applications: Currency, Fuel and Other Problems
(Std 1)

Teacher: Denis Vlismas

Exam Equivalent Time: 76.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-A2 Linear Relationships* has accounted for 5% and 4% of the 2020 and 2019 Std1 exams respectively.
- *MS-A2 Linear Relationships* is split into the two sub-topics listed in the graph above. Note that estimated exam contributions in the graph are taken from Std2/Gen2 past exams due to the very limited Std1 exam history. They should only be used for estimating the *relative* importance of each sub-topic.
- This analysis looks at *Applications: Currency, Fuel and Other Problems*.

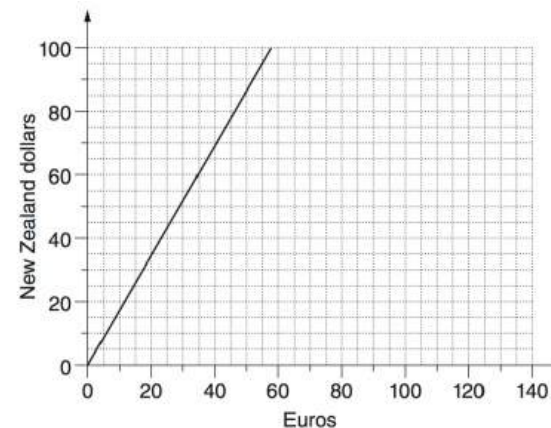
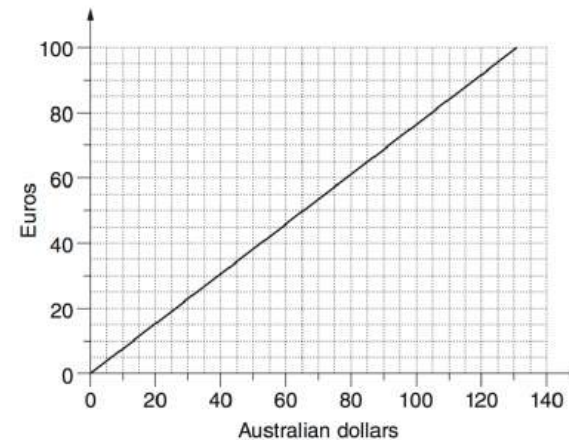
ANALYSIS - What to Expect and Common pitfalls

- Applications of *Linear Relationships* can be examined in a very broad way, as evidenced by the wide array of past HSC questions.
- Modelling a linear cost equation caused notable problems in *2020 Std1 10 MC* (mean mark 21%) and warrants attention.
- Constructing an equation of linear proportionality attracted a substantial 3-mark allocation in the *2020 Std1 exam* and also caused issues.
- *Fuel consumption* and *currency conversion* are both very common question types in this area, with the latter appearing in *2019 Std1 33* and producing sub-20% mean marks in both parts.
- Some linear modelling questions in everyday contexts, describing subjects like boot production, have seen chunky mark allocations. It is worth noting that the most important testing of this topic area is expected, in our view, to be within *MS-S3 Further Statistical Analysis* which incorporates scatterplots and lines of best fit.

Questions

1. Algebra, STD2 A2 2012 HSC 13 MC

Conversion graphs can be used to convert from one currency to another.



Sarah converted 60 Australian dollars into Euros. She then converted all of these Euros into New Zealand dollars.

How much money, in New Zealand dollars, should Sarah have?

- (A) \$26
 (B) \$45
 (C) \$78
 (D) \$135

2. Algebra, STD2 A2 2017 HSC 14 MC

Kate is comparing two different models of car. Car A uses fuel at the rate of 9 L/100 km. Car B uses 3.5 L/100 km.

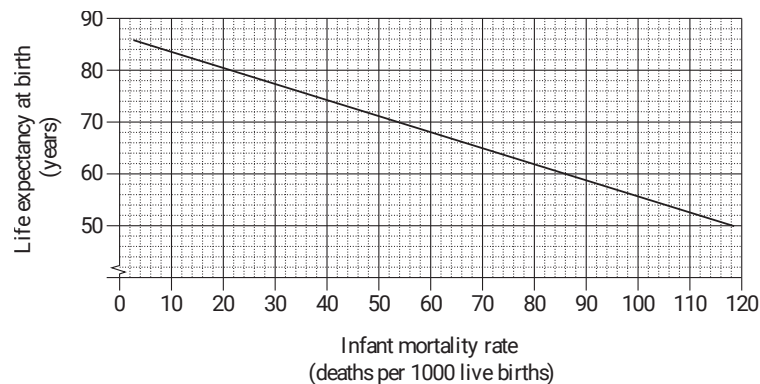
Suppose Kate plans on driving 8000 km in the next year.

How much less fuel will she use driving car B instead of car A?

- A. 280 L
- B. 440 L
- C. 720 L
- D. 1000 L

3. Algebra, STD2 A2 2017 HSC 3 MC

The graph shows the relationship between infant mortality rate (deaths per 1000 live births) and life expectancy at birth (in years) for different countries.



What is the life expectancy at birth in a country which has an infant mortality rate of 60?

- A. 68 years
- B. 69 years
- C. 86 years
- D. 88 years

4. Algebra, STD2 A2 2018 HSC 5 MC

The driving distance from Alex's home to his work is 20 km. He drives to and from work five times each week. His car uses fuel at the rate of 8 L/100 km.

How much fuel does he use driving to and from work each week?

- A. 16 L
- B. 20 L
- C. 25 L
- D. 40 L

5. Algebra, STD2 A2 SM-Bank 4 MC

A car travels 350 km on 40 L of petrol.

What is its fuel consumption?

- A. 7.8 L/100 km
- B. 8.4 L/100 km
- C. 8.8 L/100 km
- D. 11.4 L/100 km

6. Algebra, STD1 A2 2021 HSC 8 MC

A student is thinking of a number. Let the number be x .

When the student subtracts 8 from this number and multiplies the result by 3, the answer is 2 more than x .

Which equation can be used to find x ?

- A. $3(x - 8) = 2x$
- B. $3x - 8 = 2x$
- C. $3(x - 8) = x + 2$
- D. $3x - 8 = x + 2$

7. Algebra, STD2 A2 2014 HSC 22 MC

Heather's car uses fuel at the rate of 6.6 L per 100 km for long-distance driving and 8.9 L per 100 km for short-distance driving.

She used the car to make a journey of 560 km, which included 65 km of short-distance driving.

Approximately how much fuel did Heather's car use on the journey?

- (A) 37 L
- (B) 38 L
- (C) 48 L
- (D) 50 L

8. Algebra, STD2 A2 2004 HSC 22 MC

John knows that

- one Australian dollar is worth 0.62 euros
- one Vistabella dollar (**\$V**) is worth 1.44 euros.

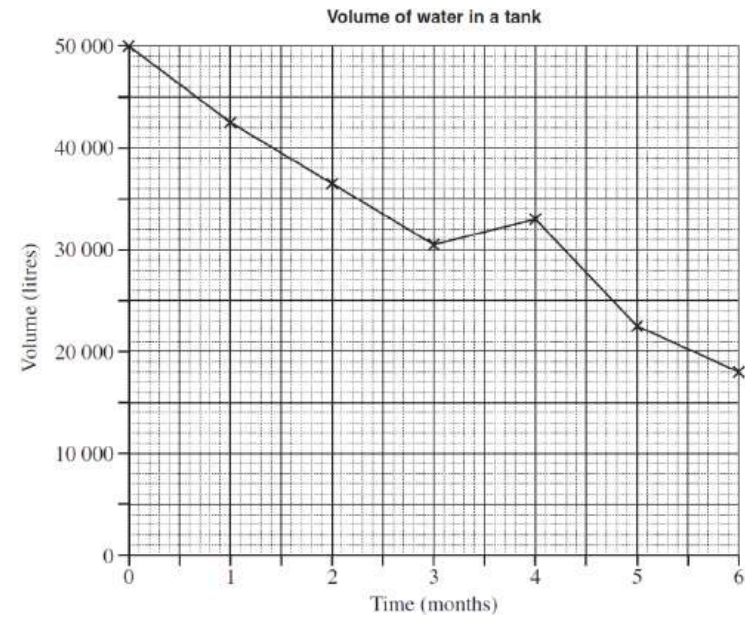
John changes 25 Australian dollars to Vistabella dollars.

How many Vistabella dollars will he get?

- (A) **\$V10.76**
 (B) **\$V22.32**
 (C) **\$V28.00**
 (D) **\$V58.06**
-

9. Algebra, STD2 A2 2009 HSC 13 MC

The volume of water in a tank changes over six months, as shown in the graph.



Consider the overall decrease in the volume of water.

What is the average percentage decrease in the volume of water per month over this time, to the nearest percent?

- (A) 6%
 (B) 11%
 (C) 32%
 (D) 64%
-

10. Algebra, STD1 A2 2020 HSC 10 MC

A plumber charges a call-out fee of \$90 as well as \$2 per minute while working.

Suppose the plumber works for t hours.

Which equation expresses the amount the plumber charges (**\$C**) as a function of time (**t** hours)?

- A. **$C = 2 + 90t$**
 B. **$C = 90 + 2t$**
 C. **$C = 120 + 90t$**
 D. **$C = 90 + 120t$**
-

11. Algebra, STD2 A2 2016 HSC 26c

Peta's car uses fuel at the rate of 5.9 L /100 km for country driving and 7.3 L /100 km for city driving. On a trip, she drives 170 km in the country and 25 km in the city.

Calculate the amount of fuel she used on this trip. **(2 marks)**

12. Algebra, STD2 A2 2014 HSC 27b

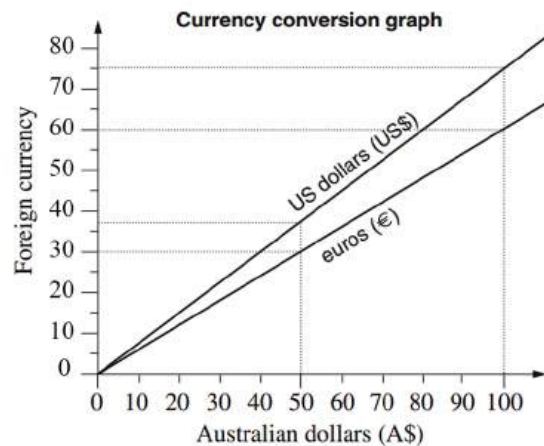
Xuso is comparing the costs of two different ways of travelling to university.

Xuso's motorcycle uses one litre of fuel for every 17 km travelled. The cost of fuel is \$1.67/L and the distance from her home to the university car park is 34 km. The cost of travelling by bus is \$36.40 for 10 single trips.

Which way of travelling is cheaper and by how much? Support your answer with calculations. **(2 marks)**

13. Algebra, STD2 A2 2007 HSC 24c

Sandy travels to Europe via the USA. She uses this graph to calculate her currency conversions.



- After leaving the USA she has US\$150 to add to the A\$600 that she plans to spend in Europe. She converts all of her money to euros. How many euros does she have to spend in Europe? **(3 marks)**
- If the value of the euro falls in comparison to the Australian dollar, what will be the effect on the gradient of the line used to convert Australian dollars to euros? **(1 mark)**

14. Algebra, STD2 A2 2014 HSC 26f

The weight of an object on the moon varies directly with its weight on Earth. An astronaut who weighs 84 kg on Earth weighs only 14 kg on the moon.

A lunar landing craft weighs 2449 kg when on the moon. Calculate the weight of this landing craft when on Earth. **(2 marks)**

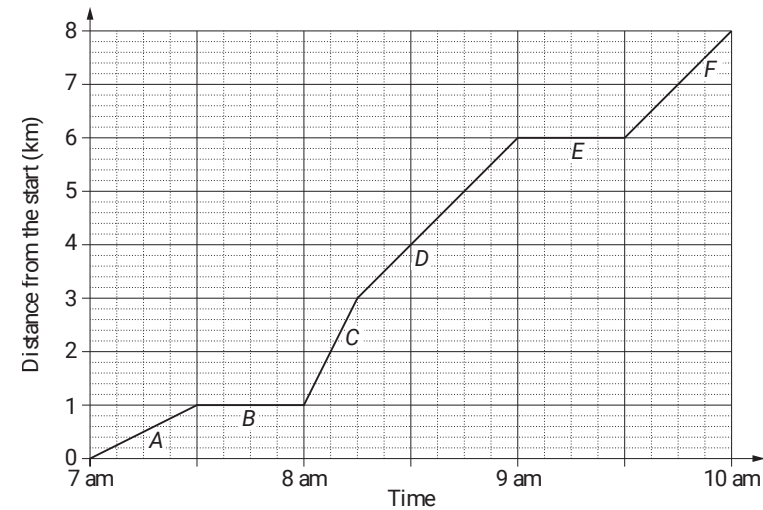
15. Algebra, STD1 A2 2021 HSC 13

The fuel consumption for a car is 6.7 litres/100 km. On a road trip, the car travels a distance of 1560 km and the fuel cost is \$1.45 per litre.

What is the total fuel cost for the trip? **(2 marks)**

16. Algebra, STD1 A2 2021 HSC 23

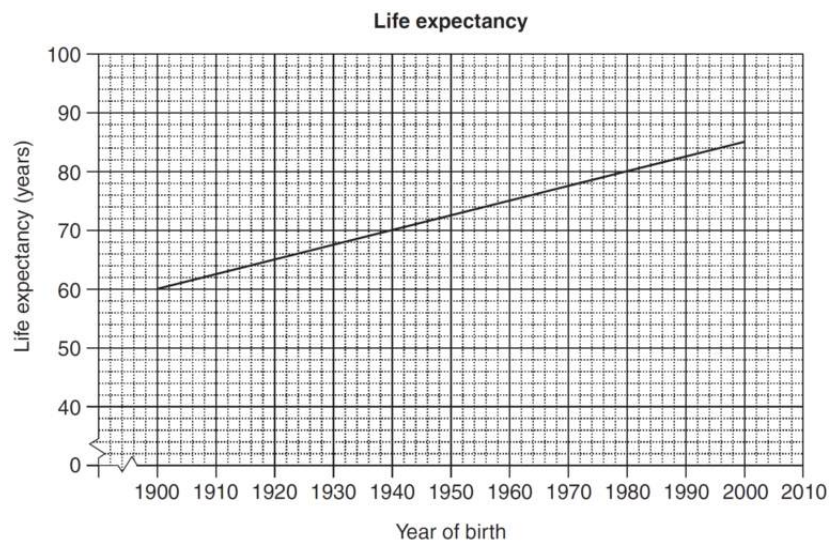
Sue walks along a trail, starting at 7 am and finishing at 10 am. The travel graph shows Sue's journey from the start to the finish. The journey has been broken into six sections, **A**, **B**, **C**, **D**, **E** and **F**.



- On two occasions Sue stopped to rest. In which sections of the journey did Sue rest? **(1 mark)**
- In which section of the journey did Sue travel fastest? Justify your answer. **(2 marks)**
- Kim walked along the same trail, also starting at 7 am and finishing at 10 am. Kim walked at a constant speed for the entire journey.
By showing Kim's journey on the grid above, determine between what times Sue was ahead of Kim. **(3 marks)**

17. Algebra, STD2 A2 2016 HSC 29e

The graph shows the life expectancy of people born between 1900 and 2000.



- According to the graph, what is the life expectancy of a person born in 1932? *(1 mark)*
- With reference to the value of the gradient, explain the meaning of the gradient in this context. *(2 marks)*

18. Algebra, STD1 A2 2020 HSC 20

The weight of a bundle of A4 paper (W kg) varies directly with the number of sheets (N) of A4 paper that the bundle contains.

This relationship is modelled by the formula $W = kN$, where k is a constant.

The weight of a bundle containing 500 sheets of A4 paper is 2.5 kilograms.

- Show that the value of k is 0.005. *(1 mark)*
- A bundle of A4 paper has a weight of 1.2 kilograms. Calculate the number of sheets of A4 paper in the bundle. *(2 marks)*

19. Algebra, STD2 A2 2007 HSC 27b*

A clubhouse uses four long-life light globes for five hours every night of the year. The purchase price of each light globe is \$6.00 and they each cost $\$d$ per hour to run.

- Write an equation for the total cost ($\$c$) of purchasing and running these four light globes for one year in terms of d . *(2 marks)*
- Find the value of d (correct to three decimal places) if the total cost of running these four light globes for one year is \$250. *(1 mark)*
- If the use of the light globes increases to ten hours per night every night of the year, does the total cost double? Justify your answer with appropriate calculations. *(1 mark)*

20. Algebra, STD2 A2 SM-Bank 3

The average height, C , in centimetres, of a girl between the ages of 6 years and 11 years can be represented by a line with equation

$$C = 6A + 79$$

where A is the age in years. For this line, the gradient is 6.

- What does this indicate about the heights of girls aged 6 to 11? *(1 mark)*
- Give ONE reason why this equation is not suitable for predicting heights of girls older than 12. *(1 mark)*

21. Algebra, STD2 A2 SM-Bank 2

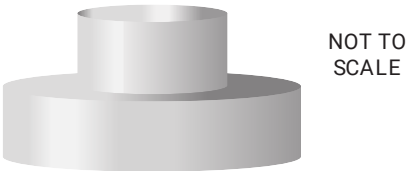
The weight of a steel beam, w , varies directly with its length, ℓ .

A 1200 mm steel beam weighs 144 kg.

Calculate the weight of a 750 mm steel beam. *(2 marks)*

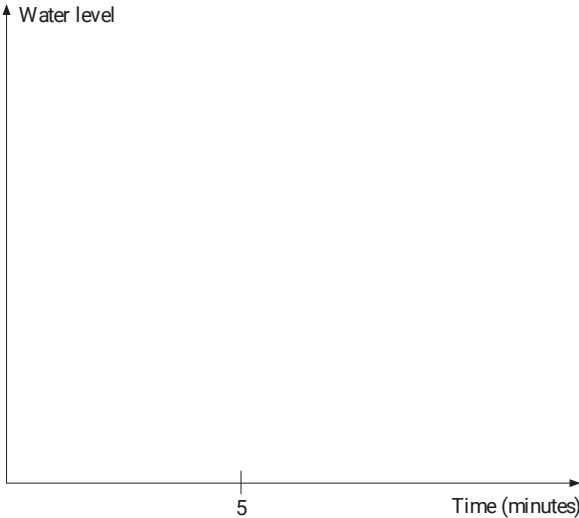
22. Algebra, STD1 A3 2021 HSC 25

The diagram shows a container which consists of a small cylinder on top of a larger cylinder.



The container is filled with water at a constant rate to the top of the smaller cylinder. It takes 5 minutes to fill the larger cylinder.

Draw a possible graph of the water level in the container against time. (2 marks)

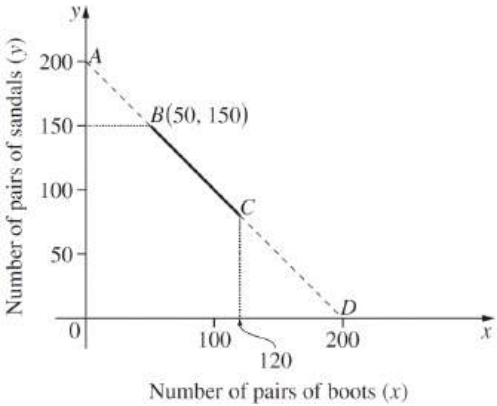


23. Algebra, STD2 A2 2009 HSC 24d

A factory makes boots and sandals. In any week

- the total number of pairs of boots and sandals that are made is 200
- the maximum number of pairs of boots made is 120
- the maximum number of pairs of sandals made is 150.

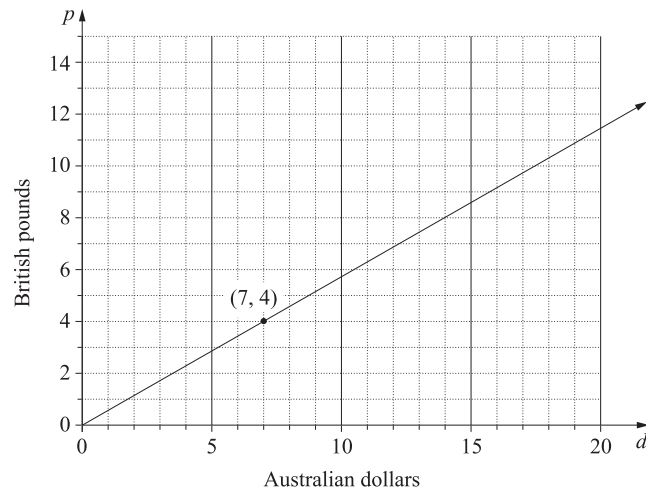
The factory manager has drawn a graph to show the numbers of pairs of boots (x) and sandals (y) that can be made.



- Find the equation of the line AD . (1 mark)
- Explain why this line is only relevant between B and C for this factory. (1 mark)
- The profit per week, $\$P$, can be found by using the equation $P = 24x + 15y$. Compare the profits at B and C . (2 marks)

24. Algebra, STD1 A2 2019 HSC 33

The relationship between British pounds (p) and Australian dollars (d) on a particular day is shown in the graph.



a. Write the direct variation equation relating British pounds to Australian dollars in the form

$p = md$. Leave m as a fraction. (1 mark)

b. The relationship between Japanese yen (y) and Australian dollars (d) on the same day is given by the equation $y = 76d$.

Convert 93 100 Japanese yen to British pounds. (2 marks)

Worked Solutions

1. Algebra, STD2 A2 2012 HSC 13 MC

Using the graphs

\$60 Australian = 46 Euro

46 Euro = \$78 New Zealand

$\Rightarrow C$

2. Algebra, STD2 A2 2017 HSC 14 MC

Fuel used by car A

$$= \frac{8000}{100} \times 9$$

$$= 720 \text{ L}$$

Fuel used by car B

$$= \frac{8000}{100} \times 3.5$$

$$= 280 \text{ L}$$

$\therefore$ Fuel saved using car B

$$= 720 - 280$$

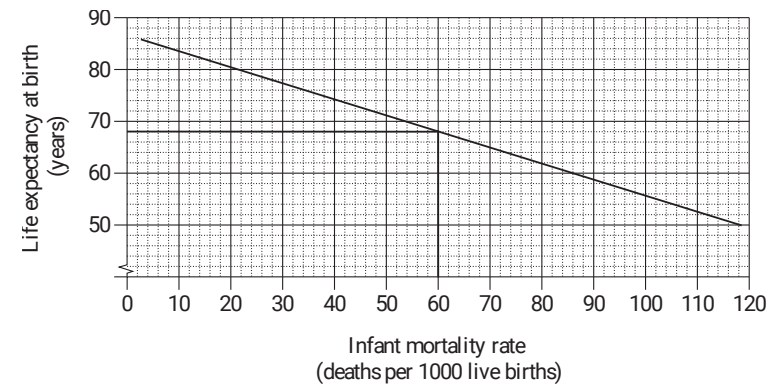
$$= 440 \text{ L}$$

$\Rightarrow B$

3. Algebra, STD2 A2 2017 HSC 3 MC

When infant mortality rate is 60, life

expectancy at birth is 68 years (see below).



$\Rightarrow A$

4. Algebra, STD2 A2 2018 HSC 5 MC

Total distance travelled each week

$$= 5 \times 2 \times 20$$

$$= 200 \text{ km}$$

$\therefore$ Total fuel used

$$= \frac{200}{100} \times 8 \text{ L}$$

$$= 16 \text{ L}$$

$\Rightarrow A$

5. Algebra, STD2 A2 SM-Bank 4 MC

40 litres are used to travel $3.5 \times 100 \text{ km}$

Fuel consumption (L/100 km)

$$= \frac{40}{3.5}$$

$$= 11.428$$

$$= 11.4 \text{ L/100 km}$$

$\Rightarrow D$

6. Algebra, STD1 A2 2021 HSC 8 MC

The description defines the following equation:

$$(x - 8) \times 3 = x + 2$$

$$3(x - 8) = x + 2$$

$\Rightarrow C$

7. Algebra, STD2 A2 2014 HSC 22 MC

Fuel used in short distance

$$= \frac{65}{100} \times 8.9 \text{ L} = 5.785 \text{ L}$$

Fuel used in long distance

$$= \frac{495}{100} \times 6.6 \text{ L} = 32.67 \text{ L}$$

$\therefore$ Total Fuel = $5.785 + 32.67$

$$= 38.455 \text{ L}$$

$\Rightarrow B$

8. Algebra, STD2 A2 2004 HSC 22 MC

John has 25 Aust dollars.

Converting to Euros

$$25 \text{ Aust} = 25 \times 0.62$$

$$= 15.5 \text{ Euros}$$

Converting to Vistabella dollars

$$15.5 \text{ Euros} = \frac{15.5}{1.44}$$

$$= \$V10.76$$

$\Rightarrow A$

9. Algebra, STD2 A2 2009 HSC 13 MC

$$\text{Initial Volume} = 50\,000 \text{ L}$$

$$\text{Final volume} = 18\,000 \text{ L}$$

$$\begin{aligned}\text{Decrease} &= 50\,000 - 18\,000 \\ &= 32\,000 \text{ L (over 6 months)}\end{aligned}$$

$$\begin{aligned}\text{Loss per month} &= \frac{32\,000}{6} \\ &= 5333.33... \text{ L per month}\end{aligned}$$

$$\begin{aligned}\% \text{ loss per month} &= \frac{5333.33...}{50\,000} \\ &= 10.666... \%\end{aligned}$$

$\Rightarrow B$

♦ Mean mark 48%
COMMENT: Remember that % decrease requires the decrease in volume to be divided by the original volume (50,000L).

10. Algebra, STD1 A2 2020 HSC 10 MC

$$\begin{aligned}\text{Hourly rate} &= 60 \times 2 \\ &= \$120\end{aligned}$$

$$\therefore C = 90 + 120t$$

♦♦♦ Mean mark 21%.

11. Algebra, STD2 A2 2016 HSC 26c

$$\begin{aligned}\text{Fuel used in country} \\ &= 170 \times \frac{5.9}{100} \\ &= 10.03 \text{ L}\end{aligned}$$

$$\begin{aligned}\text{Fuel used in city} \\ &= 25 \times \frac{7.3}{100} \\ &= 1.825 \text{ L}\end{aligned}$$

$$\begin{aligned}\therefore \text{Total fuel used} \\ &= 10.03 + 1.825 \\ &= 11.855 \text{ L}\end{aligned}$$

12. Algebra, STD2 A2 2014 HSC 27b

Compare cost of a 1-way trip

Motorcycle

$$\text{Fuel used} = \frac{34}{17} = 2 \text{ L}$$

$$\text{Cost} = 2 \times \$1.67 = \$3.34$$

Bus

$$\text{Cost} = \frac{36.40}{10} = \$3.64$$

$$\text{Difference} = \$3.64 - 3.34 = \$0.30$$

$\therefore$ Motorcycle is \$0.30 cheaper per 1-way trip.

13. Algebra, STD2 A2 2007 HSC 24c

i. From graph:

$$75 \text{ US\$} = 100 \text{ A\$}$$

$$\Rightarrow 150 \text{ US\$} = 200 \text{ A\$}$$

$\therefore$ Sandy has a total of 800 A\$

Converting A\$ to €:

$$100 \text{ A\$} = 60 \text{ €}$$

$$\begin{aligned}\therefore 800 \text{ A\$} &= 8 \times 60 \\ &= 480 \text{ €}\end{aligned}$$

ii. If the value of the euro falls against the

A\$, then 1 A\$ will buy more euros than before and the gradient used to convert the currencies will steepen (increase).

14. Algebra, STD2 A2 2014 HSC 26f

$$w_{\text{moon}} \propto w_{\text{earth}}$$

$$\Rightarrow w_{\text{m}} = k \times w_{\text{e}}$$

We know when $w_{\text{e}} = 84$, $w_{\text{m}} = 14$

$$\therefore 14 = k \times 84$$

$$k = \frac{14}{84} = \frac{1}{6}$$

If $w_{\text{m}} = 2449$ kg, find w_{e}

$$2449 = \frac{1}{6} \times w_{\text{e}}$$

$$w_{\text{e}} = 14\,694 \text{ kg}$$

$\therefore$ Landing craft weighs 14 694 kg on earth

15. Algebra, STD1 A2 2021 HSC 13

$$\begin{aligned} \text{Total fuel used} &= 6.7 \times \frac{1560}{100} \\ &= 104.52 \text{ litres} \end{aligned}$$

$$\begin{aligned} \text{Total fuel cost} &= 104.52 \times 1.45 \\ &= \$151.55 \end{aligned}$$

16. Algebra, STD1 A2 2021 HSC 23

a. *B and E*

♦ Mean mark part (b) 44%.

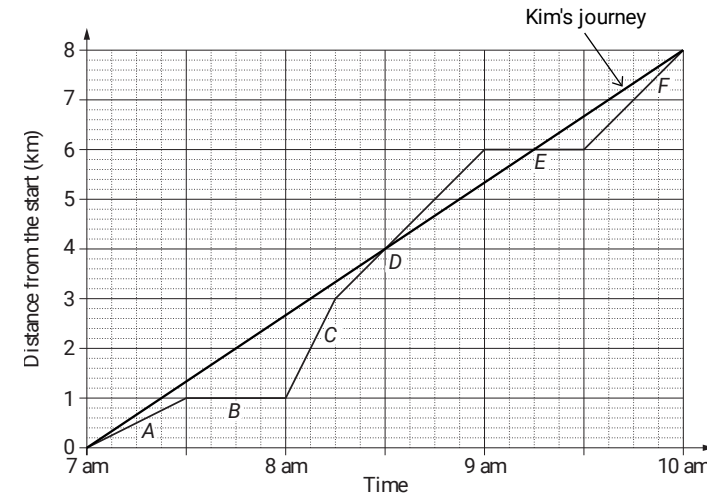
b.

Fastest travel occurs when the slope is the steepest.

$\therefore$ Section *C*

♦♦ Mean mark part (c) 30%.

c.



Sue was ahead when her graph is higher than Kim's.

$\therefore$ She was ahead between 8:30 am – 9:15 am

17. Algebra, STD2 A2 2016 HSC 29e

i. 68 years

ii. Using (1900,60), (1980,80)

$$\begin{aligned} \text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{80 - 60}{1980 - 1900} \\ &= 0.25 \end{aligned}$$

♦♦ Mean mark part (ii) 33%

$\therefore$ After 1900, life expectancy increases by 0.25 years for each year later that someone is born.

18. Algebra, STD1 A2 2020 HSC 20

a. $W = 2.5$ kg when $N = 500$:

$$2.5 = k \times 500$$

$$\therefore k = \frac{2.5}{500}$$
$$= 0.005$$

b. Find N when $W = 1.2$ kg:

$$1.2 = 0.005 \times N$$

$$\therefore N = \frac{1.2}{0.005}$$
$$= 240 \text{ sheets}$$

♦ Mean mark 50%.

19. Algebra, STD2 A2 2007 HSC 27b*

i. Purchase price = $4 \times 6 = \$24$

$$\begin{aligned}\text{Running cost} &= \# \text{ Hours} \times \text{cost per hour} \\ &= 4 \times 5 \times 365 \times d \\ &= 7300d\end{aligned}$$

$$\therefore \$c = 24 + 7300d$$

ii. Given $\$c = \250

$$\Rightarrow 250 = 24 + 7300d$$

$$7300d = 226$$

$$d = \frac{226}{7300}$$

$$= 0.03095\dots$$

$$= 0.031 \text{ \$/hr (3 d.p.)}$$

iii. If d doubles to 0.062 \\$/hr

$$\$c = 24 + 7300 \times 0.062$$

$$= \$476.60$$

Since \$476.60 is less than $2 \times \$250$ (\$500),
the total cost increases to less than double
the original cost.

20. Algebra, STD2 A2 SM-Bank 3

i. It indicates that 6-11 year old girls, on average, grow
6 cm per year.

ii. Girls eventually stop growing, and the equation doesn't
factor this in.

21. Algebra, STD2 A2 SM-Bank 2

$$w \propto \ell$$

$$w = k\ell$$

When $w = 144 \text{ kg}$, $\ell = 1200 \text{ mm}$

$$144 = k \times 1200$$

$$k = \frac{144}{1200}$$

$$= \frac{3}{25}$$

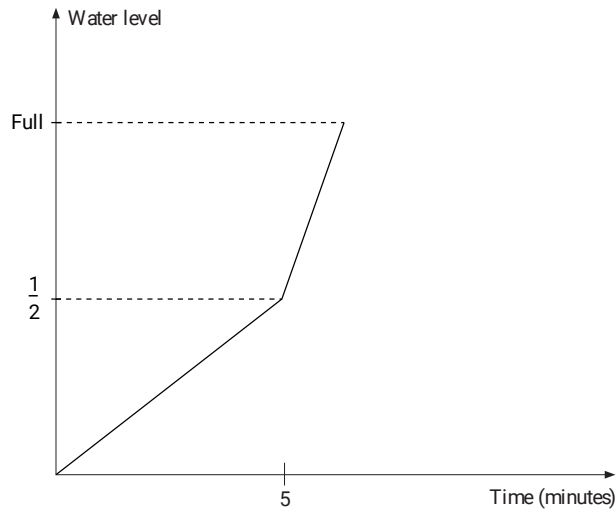
When $\ell = 750 \text{ mm}$,

$$w = \frac{3}{25} \times 750$$

$$= 90 \text{ kg}$$

22. Algebra, STD1 A3 2021 HSC 25

♦ Mean mark 38%.



23. Algebra, STD2 A2 2009 HSC 24d

i. We are told the number of boots (x),
and shoes (y), made in any week = 200
 $\Rightarrow$ Equation of AD is $x + y = 200$

♦♦♦ Mean mark part (i) 14%.
Using $y = mx + b$ is a less efficient but equally valid method, using $m = -1$ and $b = 200$ (y -intercept).

ii. Since the max amount of boots = 120

$$\Rightarrow x \text{ cannot } > 120$$

Since the max amount of sandals = 150

$$\Rightarrow y \text{ cannot } > 150$$

$\therefore$ The line AD is only possible between B and C .

♦ Mean mark 49%

iii. At B , $x = 50$, $y = 150$

$$\Rightarrow \$P \text{ (at } B) = 24 \times 50 + 15 \times 150$$

$$= 1200 + 2250$$

$$= \$3450$$

At C , $x = 120$, $y = 80$

$$\Rightarrow \$P \text{ (at } C) = 24 \times 120 + 15 \times 80$$

$$= 2880 + 1200$$

$$= \$4080$$

♦ Mean mark 40%.

$\therefore$ The profits at C are \$630 more than at B .

24. Algebra, STD1 A2 2019 HSC 33

a. $m = \frac{\text{rise}}{\text{run}} = \frac{4}{7}$

$$p = \frac{4}{7}d$$

◆◆◆ Mean mark part (a) 8%.

b. Yen to Australian dollars:

$$y = 76d$$
$$93\,100 = 76d$$
$$d = \frac{93\,100}{76}$$
$$= 1225$$

◆◆ Mean mark part (b) 16%.

Aust dollars to pounds:

$$p = \frac{4}{7} \times 1225$$
$$= 700 \text{ pounds}$$

$$\therefore 93\,100 \text{ Yen} = 700 \text{ pounds}$$